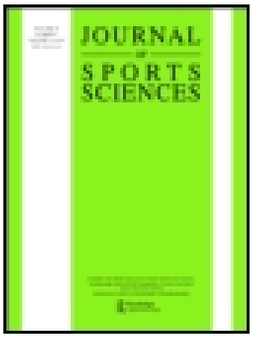

Journal of Sports Sciences

ISSN: 0264-0414 (Print) 1466-447X (Online) Journal homepage: http://www.tandfonline.com/loi/rjsp20

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To cite this article: Ross H. Sanders (2007) Kinematics, coordination, variability, and biological noise in the prone flutter kick at different levels of a "learn-to-swim" programme, Journal of Sports Sciences, 25:2, 213-227, DOI: 10.1080/02640410600631025

To link to this article: http://dx.doi.org/10.1080/02640410600631025

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Journal of Sports Sciences, January 15th 2007; 25(2): 213–227

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Kinematics, coordination, variability, and biological noise in the prone flutter kick at different levels of a ''learn-to-swim'' programme

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(Accepted 6 February 2006)

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Abstract

The purpose of this study was to establish the characteristics of the movement patterns common to flutter kicking of skilled swimmers and to determine how the movement patterns of swimmers at different levels of a ''learn-to-swim'' programme differ from those of skilled swimmers. Also, the nature of the skill afforded the opportunity to investigate learning of a cyclical multi-joint task from a motor control perspective. The underwater motion of nine children representing three levels of a ''learn-to-swim'' programme and 10 skilled swimmers were video-recorded while performing nine cycles of prone flutter kicking. Kinematics including joint angular motion and coordination of joint actions were calculated. Fourier analysis was applied to determine the frequency composition of the vertical undulations of the hip, knee, and ankle and to calculate the velocity of the body wave travelling caudally from hip to ankle. Fourier analysis also enabled investigation of biological noise, as distinct from variability. The results indicated the desired joint angles and coordination towards which learners could be guided. An index based on the ratio of hip–knee and knee–ankle body wave velocities showed that the inter-joint coordination of most learners was not appropriate for effective flutter kicking. There was strong evidence to suggest that skilled performance in flutter kicking is characterized by sequencing of joint actions to produce a single sinusoidal body wave moving caudally with not decreasing and preferably increasing velocity, low biological noise, and small variability.

Keywords: *Biological noise, coordination, dynamical systems, flutter kick, Fourier analysis, skill acquisition, swimming, variability*

Introduction

A coach, teacher, or instructor interacts with learners to develop skills. One objective is to guide the learner towards movement patterns that optimize performance. An effective instructor assesses the level of a learner relative to the scope of skill development, identifies key aspects of movement that determine success, and decides when intervention is necessary to modify movement patterns or progress to the next stage of skill development (Knudson & Morrison, 1997). These tasks demand knowledge of what constitutes skilled performance as well as the ways in which performance at each level of a ''learn-to-swim'' programme may differ from skilled performance. Fundamental to the process of effective instruction is the ability to recognize that individuals differ in the way they deviate from movement patterns that are characteristic of skilled performance. For example, Halverson and Williams (1985) have shown that children learning to hop may achieve advanced coordination patterns with common kinematic characteristics despite displaying very different initial kinematics and paths of progression.

The reduction in differences among individuals as they become more skilled is a commonly observed phenomenon (Boyle & Ackerman, 2004). These characteristics may be embodied in variables that vary within narrow ranges among skilled performers relative to the ranges exhibited by learners. Examples include the range of joint angular motion and the temporal relationships among joint motions. Many comparisons of kinematics across skill levels in multiple-joint sports skills have been conducted (Anderson & Sidaway, 1994; Goodway, Rudisill, & Valentini, 2002; Langendorfer & Roberton, 2002; Temprado, Della-Grasta, Farrell, & Laurent, 1997). In general, research has shown that as practice and learning progressed, the pattern of coordination more closely resembled the patterns of experienced performers. While it is well established that movement patterns converge as learners' skill increases,

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ISSN 0264-0414 print/ISSN 1466-447X online © 2007 Taylor & Francis
DOI: 10.1080/02640410600631025

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instructors need to know what the characteristics of the skilled performance are and how the movement patterns of learners at different levels of a "learn-to-swim" programme commonly differ from the movement patterns of skilled performers. Therefore, the characteristics of kinematic variables representing observable features of a skill, and which collectively define the movement patterns of skilled performers, provide important "benchmarks" towards which instructors can guide learners.

For safety reasons, swimming is an important skill for children to acquire at a young age. Thus, water safety and water confidence have been prominent themes in research of learning to swim. Ramsey (1980) and Blanksby, Parker, Bradley and Ong (1995) have commented that, in general, the development of skill in the formal strokes has not been a priority among researchers. However, because well-established coordination patterns are not necessarily the most appropriate for good performance (Whitall, 2003) and are known to be difficult to change in swimming skills (Corlett, 1980; Costill, Maglischo, & Richardson, 1992), the process of developing appropriate movement patterns should begin from early in the development of the skill.

Some research has been conducted on coordination of children in aquatic environments. Wielki and Houben (1983) described the movement patterns of infants from analysis of underwater video recordings. However, this was from a developmental and water safety perspective rather than from the perspective of acquisition of skill in the formal swimming strokes. Other researchers of children in aquatic environments have focused on pedagogical aspects to find new methods of teaching swimming skills to children (Killian, 1988). Although Langendorfer (1986) highlighted the need for aquatic research applied to children and a greater understanding of motor skill acquisition in children learning to swim, there remains limited data on the acquisition of skill in specific swimming tasks, particularly the flutter kick used in freestyle swimming. However, Watkins and Gordon (1983), Hollander, de Groot, van Ingen Schenau, Kahman and Toussaint (1988), and Adams (2000) have indicated that the freestyle kick does contribute to propulsion. Furthermore, the kick plays an important role in providing a stable "platform" for the whole stroke by facilitating body position to optimize propulsion and minimize resistance, and in assisting an economical body roll (Maglischo, 1993). In "learn-to-swim" programmes, it is common to teach kicking in the prone position in the belief that proper kicking is required as a foundation for development of good coordination in the entire freestyle stroke.

The purpose of this study was to establish the characteristics of the movement patterns common to skilled flutter kicking in the prone position and to determine how the movement patterns of swimmers at various levels of a "learn-to-swim" programme differ from those of skilled swimmers. This would yield information to help instructors guide learners towards good technique in flutter kicking in freestyle swimming.

Although the experiment was designed to yield information to aid teaching of the flutter kick rather than to test specific hypotheses in motor control, the data are well suited to scrutiny with a view to contributing to the "jigsaw" of information relating to current issues in motor control. These issues include development and stabilization of new movement patterns, variability, biological noise, and reduction in complexity of control.

From a motor control perspective, the flutter kick is a cyclical multi-joint task with strong environmental and within-participant constraints guiding the movement pattern towards a solution that has known essential characteristics. These essential characteristics arise from the interaction of the intrinsic mechanical characteristics of the human system with a fluid environment to achieve effective and physiologically efficient propulsion. It is known from studies of aquatic locomotion of various marine organisms (for a review, see Sfakiotakis, Lane, & Davies, 1999) and humans (Sanders, Cappaert, & Devlin, 1995; Ungerechts, 1982, 1985) that undulatory motion is effective in producing propulsion. Sequential displacement of body parts yields a sinusoidal wave progressing caudally along the body with increasing amplitude. The waveform is simple, comprising just a single frequency. There is no need for the movement pattern to comprise higher frequency waveforms superposed on the fundamental waveform. Logically, higher frequency waveforms would do work on the water that was not conducive to increased propulsion, would reduce the propulsive effect of the fundamental waveform, and would incur increased physiological cost.

Newell and Corcos (1993) have indicated that variability in measures relating to goal achievement often includes "biological noise" that limits performance and should therefore be eliminated. In this task, performance is related to the behaviour of a wave comprising a single sinusoid. Any waveforms superposed on that fundamental waveform represent noise. If the high-frequency waveforms associated with random error in the data collection process are removed, then the frequencies intermediate between the fundamental frequency and those associated with measurement error can be regarded as biological noise. Fourier analysis, a method of quantifying the amplitude and phase of waveforms (integer harmonics) comprising any time-series data, can be used to remove the high-frequency waveforms associated with measurement error, and quantify the power

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contained in the useful waveform and the biological noise. This process enables assessment of how the magnitude of biological noise in flutter kicking changes across levels of the "learn-to-swim" programme. Fourier analysis is presented in many mathematics and engineering texts and in-depth treatments are also available (e.g. Bloomfield, 1976; Jenkins & Watts, 1968).

Another essential characteristic of the waveform to achieve good performance in the flutter kick is the sequencing of the undulations of the body parts so that the wave moves along the body in a consistent manner. Two main mechanisms of force production by undulation have been identified (Sfakiotakis *et al.*, 1999). These are the effects of accelerating water backwards (added mass effects) and lift obtained from rotating water masses (vortex effects). Added mass effects predominate in long, slender "anguilliform" marine animals such as eels that transmit undulations along their length. Vortex effects predominate in "thunniform" marine animals such as dolphins that have relatively stiff upper bodies and large caudal fins and so have undulations confined to the caudal third of the body. It is not known which mechanism predominates in the human flutter kick but the efficiency of propulsion from vortex mechanisms is limited by the small surface area of the feet relative to the caudal fin of thunniform swimmers (Ungerechts, 1985). However, regardless of the mechanism, it is apparent that the wave should progress with increasing amplitude and in a manner that avoids sudden changes in wave velocity. Ungerechts proposed that the motion in the human double (dolphin) kick used in butterfly swimming should be "whip-like" to be effective. Indeed, elite butterfly swimmers are characterized by an increasing wave amplitude and faster progression from knee to ankle than from hip to knee (Sanders *et al.*, 1995). Therefore, it is possible that similar wave characteristics are features of skilled performance in the flutter kick. If this is the case, then the opportunity arises to investigate differences between skilled and less skilled flutter kicking with reference to the coordination of actions to produce a body wave with the desired velocity characteristics.

To achieve the desired characteristics of wave motion, swimmers must learn to sequence the joint actions appropriately. One would anticipate that shifts from established movement patterns such as those developed for ambulation (Thelen, 1995) would be required. To enable this change towards a movement pattern suited to the task, one might anticipate a freeing of degrees of freedom to facilitate exploration of the "state space" (Kugler, Kelso, & Turvey, 1980, 1982; Thelen, 1995). This would be manifest by large variability in timing of joint actions during that period. After a new pattern is established that is appropriate for the task, one might expect it to become a stable "attractor state" (Kelso, Holt, Rubin, & Kugler, 1981) characterized by small variability. Furthermore, to simplify the task, one might expect couplings between joints to eliminate "redundant" degrees of freedom. This would enable swimmers to apply the same movement pattern but with varying amplitudes and frequencies to adjust to changing demands of speed and sustainability. Coupling would be manifest in high correlations between adjacent joints of the joint angular ranges of motion for individual swimmers.

Although logical with respect to the requirements of this task, freeing and then freezing degrees of freedom is counter to the idea that degrees of freedom are initially "frozen" and then "freed" (Bernstein, 1967). Bernstein's theory is well known and is supported by empirical evidence, for example in learning a ski simulator task (Vereijken, van Emmerick, Whiting, & Newell, 1992). However, a review by Newell and Vaillancourt (2001) indicated that the issue is not yet resolved and may be task-dependent. In a dynamic balancing task (Ko, Challis, & Newell, 2003), participants initially exhibited large ranges of angular motion at all joints with high variability and low correlations between joints. Then, rather than freeing the joint motions with practice, the joint motions were constrained in amplitude. Strong correlations indicated the development of couplings between joints.

Thus, in addition to providing information useful for swimming instruction, the flutter kick was investigated with a view to contributing to the pool of information relating to contemporary motor control issues, including inter-joint coordination, variability, and biological noise.

Methods

Participants

To establish the movement patterns common to the prone flutter kicks of skilled swimmers, 10 adults (six males and four females) were recruited to form the skilled swimmer group (SKS1 to SKS10). The mean physical characteristics of these participants were as follows: age 28.9 years, body mass 76.9 kg, height 1.77 m. Their mean previous swimming experience was 17.3 years and each had a minimum of 5 years of competitive swimming experience.

Nine children aged between 4 and 11 years were recruited from the "learn-to-swim" programme of a private learn-to-swim company, Swimming Nature Limited, to determine how the movement patterns of swimmers at various levels of a "learn-to-swim" programme differ from skilled performers. Their ages, swimming level, and gender are shown in Table I. Level 1 swimmers (L1S1 to L1S3) were able

Table I. Characteristics of the participants in each of the levels of the "learn-to-swim" programme.

Level	Age (years)	Sex	Height (m)	Body mass (kg)
1	7	F	1.37	34.2
1	6	M	1.27	28.6
1	4	M	1.12	22.5
2	10	F	1.27	23.6
2	7	M	1.22	24.4
2	8	M	1.42	38.5
3	9	F	1.36	35.8
3	11	F	1.52	43.2
3	10	M	1.55	42.4

to float independently on their front and back as well as achieve propulsion through kicking. Level 2 swimmers (L2S1 to L2S3) were able to perform the full freestyle and backstroke motions for 15 m. Swimmers at Level 3 (L3S1 to L3S3) were able to bilaterally breathe (breathe to either the right or the left sides) during freestyle.

Apparatus

A video camera (KY32 CCD, JVC Corporation, Japan) in a waterproof housing recorded the underwater motion of the swimmers at 50 fields per second. The camera was positioned perpendicular to the line of the swimmer's motion. To reduce errors due to distortion, the camera was immersed in the water rather than viewing through a periscope system. Low distortion and good reliability were established using the same camera system by Psycharakis, Sanders and Mill (2005). The camera capture area was adjusted so that the entire swimmer was recorded for a minimum of three kicking cycles. A kicking cycle was defined as the period from the lowest position of the foot marker following the downbeat to the equivalent position in the next cycle.

A scale line, with markers positioned every 1 m, was placed across the pool in the direction of the line of swimming. The scale line was placed as close to the swimmer as possible, and at a depth sufficient to avoid interference with the kicking motion. To minimize scaling errors, the swimmers were instructed to swim with either their right or left side of their body, dependent on direction of swimming, directly above the scale line.

Procedure and instructions

All adult participants and the parents of the swimmers who were minors signed informed consent forms in accordance with the university's ethics protocols. Before recording, black waterproof markers were placed on anatomical landmarks on the skin surface such that an imaginary line from the camera through the marker would pass through the joint axis of rotation. Participants were asked to flex and extend the joint to assist in identifying these points while the axis was palpated by the researcher. The process was aided by identifying landmarks known to correspond closely with joint axes of rotation. These were the greater trochanter for the hips, lateral condyle of the femur for the knees, and lateral malleolus for the ankles. To estimate the camera to joint axis line for the shoulders, the participants emulated the prone swimming position with the arms extended in a streamlined position. The foot markers were placed on the lateral tuberosity of the fifth metatarsal joint.

After marker placement, each swimmer was asked to flutter kick in water 1 m deep at a "comfortable" pace for 10 m. Participants maintained a prone position with arms extended forward close to the surface and parallel to it. Ten trials were collected, five for each of the right and left sides. Trials 2, 3, and 4 of each side of each participant were used for subsequent analysis. This allowed for the possibility that the first and last trials may not be typical due to lack of familiarity or over-arousal in the case of the first trial and distracting thoughts about "what comes next" in the case of the last trial. The child participants were all at the end of their respective "learn-to-swim" programmes and so were able to assess a comfortable pace that could be sustained across the ten trials. Similarly, the adult participants could assess the pace that could be sustained comfortably across the ten trials. Adequate rest was provided between trials to ensure that fatigue did not affect the consistency of effort or level of performance. The researchers ensured that swimmers did not try to swim at maximum pace and that they had sufficient time to recover between trials. The data acquired from each trial consisted of three consecutive kicking cycles in which the lower limbs were most close to the centre of the field of view. These were subsequently separated into individual cycles. In total, 18 kicking cycles, nine left and nine right, were analysed for each participant.

Digitizing and filtering

After the video recording sessions, the landmarks were digitized using an APAS System (Ariel Dynamics Ltd, CA USA). The black markers enabled the APAS system to track and digitize the landmarks automatically to yield two-dimensional positional data. Automatic digitizing ensured that digitizing errors were minimized and that marker location was extremely consistent across video fields and trials. These data were then input to a custom-designed program in MATLAB (The MathWorks, MA USA) to calculate the variables of interest.

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In addition to the three complete kicking cycles included from each trial, 20 frames on either side of the three cycles were digitized to minimize cycle endpoint errors due to the filtering process. All data were filtered using a second-order dual-pass Butterworth filter with a cut-off frequency of 6 Hz. For comparison of timing occurrences within the kicking cycle, data sets were interpolated and normalized to percentiles of the kicking cycle (0 – 100%). This ensured that comparisons of movement patterns could be made regardless of the frequency of the kicking cycle.

Variables analysed

The study focused on key aspects of kicking including hip, knee, and ankle joint angle minima, maxima, range of motion, coordination, biological noise in the vertical undulations of the joints, and the between-participant and within-participant variability.

In this study, "coordination" was defined as "the generation of appropriate spatial and temporal relations among movement-related events such that the goal of the action is successfully achieved" (Walter, Swinnen, Corcos, Pollatou, & Pan, 1997, cited in Walter, 1998). Thus, the spatial and temporal relations of the joint linear and angular motions of the skilled swimmers were used as indicators of what was appropriate for achieving the goal while also bearing in mind the scientific rationale described in the Introduction. In particular, this task required a sequencing of joint actions such that a body wave travelled caudally. Thus, coordination was indicated by: the time percentile of attainment of minimum and maximum angles of the hip, knee, and ankle; the phase differences of the fundamental Fourier waveform representing the vertical undulation of the shoulder, hip, knee, and ankle; and the associated velocity of the "wave" that moved along the body as a result of cephalo-caudal sequencing of the joint vertical undulations (Sanders *et al.*, 1995). Coordination of the joints in a "simultaneous" fashion, which may be characteristic of unskilled performers (Kreighbaum & Barthels, 1990), would be manifest in high wave velocities. Conversely, distinct sequencing of joint actions that may be characteristic of skilled performers would be manifest in moderate wave velocities.

Calculation of variables

Joint angles were calculated as the difference in angles of the lines from the joint centre to those of adjacent joints. The angles of each line were determined using the arctangent of the ratio of differences in y and x coordinates with the standard corrections for quadrant and with flexion angles being less than 180°.

Fourier phase angles and body wave velocity were calculated using the method presented by Sanders *et al.* (1995). To quantify biological noise, the amplitude of the Fourier sine and cosine coefficients were calculated with the following equation:

$$C_n = \left(A_n{}^2 + B_n{}^2\right)^{1/2}$$

where A_n and B_n are the sine and cosine coefficient respectively for the nth Fourier harmonic.

The average power of each harmonic was calculated as $2C_n{}^2$ and then expressed as a percentage of the total of the power contained in all harmonics comprising the signal. The percentage of power in the frequency harmonics other than the fundamental Fourier frequency was regarded as the measure of biological noise.

Normalization

An important consideration was to ensure that comparisons were valid despite large between-participant differences in physical characteristics such as limb lengths and inertias and, particularly in the case of variables used to address the motor control issues, to allow comparisons of movement patterns independent of amplitude and frequency of the kick.

Time normalization to percentiles of the cycle provided an equivalent time base for comparisons of movement patterns in terms of the joint linear and angular displacements, and timing of events, especially maxima and minima of joint linear and angular displacements. This was achieved by applying the Fourier functions to output 101 points representing the kicking cycle in the inverse transform.

Importantly, the variables measured to investigate the motor control issues were independent of time with the exception of wave velocity, as this was a time derivative. Also, the comparisons of ranges of motions were made in terms of joint angles and were therefore independent of participants' physical characteristics. The Fourier power distribution used to obtain measures of smoothness and noise was expressed as a percentage of power in the whole signal and was therefore independent of both time and overall amplitude of vertical movement. Thus, valid comparisons of spatial movement variables related to the task could be made across participants and across trials within participants without concern for differences in kicking rate, limb lengths, and inertias.

Phase differences between joints were expressed as a fraction of the cycle by dividing by 360°. Phase differences appropriate for goal achievement could be expected to vary slightly across participants due to differences in the lengths of the limbs relative to each

other. However, in keeping with the rationale outlined in the Introduction, the velocity of wave travel from knee to ankle relative to that from hip to knee is related to efficiency and effectiveness of the kick and the inter-joint coordination of the kick should be adjusted towards the optimal pattern of velocity regardless of relative limb lengths. Therefore, a meaningful time-independent dimensionless index was defined as the ratio of hip – knee and ankle – knee velocity.

Statistical analysis

The variables analysed statistically included: the maximum, minimum, and range of motion of the hip, knee, and ankle joints; the time of occurrence of maximum and minimum joint angle as a percentage of the cycle; differences between joints in Fourier phase angle; body wave velocities and body wave index; and biological noise. For each of the three learning groups, the skilled group, and a pooled learning group comprising the three learning groups, mean and standard deviations between participants were determined. The standard deviations between participants provided a measure of between-participant variability for each group. Standard deviations across trials within participants were determined as scores of within-participant variability. To provide a representative score for each group, these were averaged across the members of the groups. To test for differences in means and in within-participant variability between the skilled and the pooled learning groups, two-tailed *t*-tests for groups with unequal variances were used. This test enabled valid and conservative testing of differences despite the difference in magnitude of variances of the two groups. Pearson product – moment correlations were used to assess the interdependence of ranges of motion of adjacent joints across trials within participants for both the right and left sides.

Results

Joint angular data

Table II presents the joint angular data for each of hip, knee, and ankle joints. Within-participant correlations between the range of motion of the ankle and knee (Figure 1a) indicated that there was no relation between ankle and knee range of motion among the learning swimmers with the exception of the left ankle and knee of L1S1 and the right ankle and knee of L3S2 ($P < 0.05$). The pattern appears random with the mean correlation being very small (mean $r = 0.042$).

Among the skilled group there were five significant positive correlations and one significant negative correlation. However, the mean (mean $r = 0.390$)

Table II. Mean angles (degrees) of minimum (Min.), maximum (Max.), and range of motion (ROM) of the hip, knee, and ankle joints with between-participant standard deviations (in parentheses), and mean within-participant variability (Var.) for Level 1 (L1), Level 2 (L2), Level 3 (L3), pooled learning groups (LP), and skilled (SK) groups.

		L1	L2	L3	LP	SK	*P*
Hip							
Min.	Mean	157.0 (10.21)	159.2 (18.31)	167.3 (7.75)	161.1 (12.95)	164.5 (6.52)	0.34
	Var.	8.70	5.28	6.10	6.69	2.90	**
Max.	Mean	182.0 (7.67)	182.7 (8.47)	184.0 (3.47)	182.9 (6.33)	189.8 (5.34)	***
	Var.	6.69	5.61	4.71	5.67	1.99	***
ROM	Mean	25.01 (13.29)	23.5 (11.75)	16.7 (6.83)	21.8 (10.97)	25.3 (5.41)	0.22
	Var.	7.21	4.26	4.28	5.25	2.60	***
Knee							
Min.	Mean	101.1 (14.15)	113.3 (30.67)	126.8 (7.60)	113.8 (21.66)	120.3 (8.90)	0.24
	Var.	12.65	5.97	4.60	7.74	3.89	**
Max.	Mean	173.5 (12.73)	175.4 (6.73)	177.8 (5.59)	175.6 (8.57)	179.7 (4.24)	0.077
	Var.	4.99	2.74	1.86	3.20	1.36	***
ROM	Mean	72.4 (21.60)	62.1 (29.61)	51.0 (11.80)	61.8 (22.74)	59.4 (9.22)	0.67
	Var.	11.91	6.41	5.23	7.85	4.20	**
Ankle							
Min.	Mean	137.2 (6.66)	136.0 (12.03)	146.8 (2.42)	140.0 (9.05)	152.9 (4.38)	***
	Var.	10.70	8.04	5.28	8.01	2.53	***
Max.	Mean	166.7 (8.49)	167.3 (5.24)	170.5 (3.64)	168.1 (6.01)	173.3 (6.66)	*
	Var.	9.80	6.26	4.70	6.92	2.07	**
ROM	Mean	29.5 (5.36)	31.2 (11.54)	23.7 (3.86)	28.1 (7.93)	20.4 (5.14)	**
	Var.	8.27	7.82	7.08	7.72	3.14	***

Note: *P*-values are for two-tailed *t*-tests for groups with unequal variances comparing the SK and LP groups for means and within-participant variability. $*P < 0.05$; $**P < 0.01$; $***P < 0.001$.

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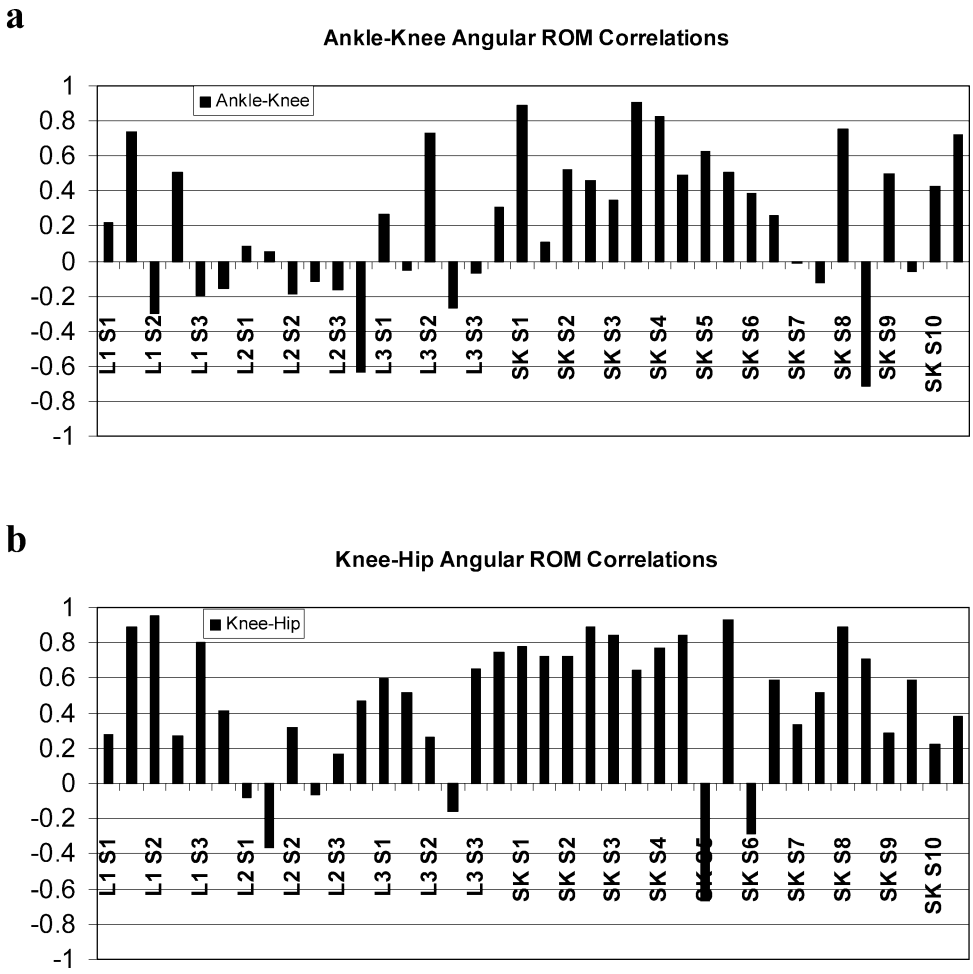

Figure 1. Correlations between joint ranges of motion (ROM) for (a) ankle and knee and (b) knee and hip.

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indicated that overall there was an association between the ranges of motion of the ankle and knee that, while weak, suggests some interdependence. The difference in correlations between the pooled learner and skilled groups was significant ($P < 0.01$).

Correlations between the range of motion of the knee and hip (Figure 1b) were predominantly positive for both learners (mean $r = 0.37$) and skilled swimmers (mean $r = 0.54$). The difference between the correlations of the two groups was not significant ($P = 0.20$). Among the learners five (of 18) were statistically significant ($P < 0.05$), while among the skilled group 10 (of 20) were significant ($P < 0.05$). For five of the skilled swimmers, the correlations were strong across both left and right sides, whereas none of the learners had strong correlations across both sides.

Coordination

Table III gives the mean time (percent of the cycle) of occurrence of minimum and maximum joint angles. The time of occurrence of the maximum corresponds to the time of transition from extension into flexion, while the time of occurrence of the minimum corresponds to the time of transition from flexion into extension.

Table IV lists the mean wave velocity of the fundamental Fourier frequency from hip to knee and from knee to ankle joint of the combined right and left sides with mean standard deviations across all trials of all participants in the group. The group descriptive statistics for velocity of the wave indicated that, in general, the hip and knee vertical undulations were much more simultaneous for the learning groups than for the skilled swimmers. However, the large between-participant standard deviations indicated marked differences between the swimmers in the learning groups. Figure 2 provides detailed information about the Fourier phase differences between joints (Figure 2a–c) and the velocity index (Figure 2d) of individuals and clearly indicates the between-participant variability and the within-participant variability.

The difference in phase between the skilled group and the combined learning groups was significant ($P < 0.01$). However, the variability among

Table III. Time (percent of cycle) of occurrence of minimum (Min.) and maximum (Max.) hip, knee, and ankle joint angles with between-participant standard deviations (in parentheses), and mean within-participant variability (Var.) for level 1 (L1), level 2 (L2), level 3 (L3), pooled learning groups (LP), and skilled groups (SK).

		L1	L2	L3	LP	SK	*P*
Hip							
Min.	Mean	61.9 (13.51)	65.9 (12.69)	83.0 (1.52)	70.2 (13.81)	82.2 (2.67)	**
	Var.	21.13	18.49	8.39	16.84	1.83	***
Max.	Mean	30.7 (8.96)	27.0 (16.96)	33.7 (3.09)	30.5 (10.91)	40.6 (2.28)	**
	Var.	16.27	5.95	7.92	10.05	2.09	***
Knee							
Min.	Mean	60.2 (4.68)	58.7 (5.05)	61.4 (2.94)	60.1 (4.21)	66.2 (2.16)	***
	Var.	5.31	2.69	5.01	4.34	1.60	**
Max.	Mean	7.0 (6.31)	6.1 (4.62)	7.0 (2.74)	6.7 (4.51)	9.4 (2.17)	*
	Var.	8.04	2.37	3.44	4.62	1.56	0.063
Ankle							
Min.	Mean	26.2 (8.65)	23.9 (11.07)	16.3 (4.96)	22.1 (9.18)	34.9 (10.29)	***
	Var.	14.66	14.22	10.92	13.27	10.36	0.17
Max.	Mean	72.4 (7.12)	72.6 (6.92)	73.3 (3.45)	72.8 (5.71)	73.8 (3.86)	0.51
	Var.	15.21	9.13	7.64	10.66	3.77	**
Hip – knee							
Min.	Mean	1.6 (10.03)	7.1 (10.45)	21.6 (4.14)	10.1 (11.92)	16.0 (3.23)	0.056
	Var.	20.56	19.15	9.36	16.36	2.08	**
Max.	Mean	23.7 (11.82)	20.9 (14.66)	26.7 (2.02)	23.8 (10.56)	31.3 (3.87)	**
	Var.	20.54	6.00	8.05	11.53	2.20	***
Knee – ankle							
Min.	Mean	34.0 (4.83)	34.9 (9.52)	45.1 (5.67)	38.0 (8.35)	32.4 (9.77)	0.062
	Var.	15.09	14.69	11.44	13.74	10.18	0.088
Max.	Mean	34.6 (8.92)	33.5 (8.86)	33.7 (1.86)	33.9 (6.91)	35.5 (4.34)	0.40
	Var.	17.74	9.12	7.75	11.53	3.92	**

Note: *P*-values are for two-tailed *t*-tests for groups with unequal variances comparing the SK and LP groups for means and within-participant variability. $*P < 0.05$; $**P < 0.01$; $***P < 0.001$.

Table IV. Mean wave velocities ($m \cdot s^{-1}$) and standard deviations (in parentheses).

	Hip – knee velocity		Knee – ankle velocity	
Level 1	8.2	(7.1)	2.5	(0.5)
Level 2	8.3	(3.6)	4.1	(0.5)
Level 3	7.3	(4.3)	3.8	(0.8)
Skilled	2.8	(0.5)	3.2	(0.3)

participants in the phase differences between the hip and ankle Fourier fundamental frequency was modest (Figure 2a). Furthermore, while the within-participant variability of the learner groups was generally larger than that of the skilled participants ($P < 0.001$), in many cases the within-participant variability was small relative to the within-participant variability in hip – knee phase differences (Figure 2b). Figure 2b indicates that participants in all learning groups had smaller hip – knee phase differences than the skilled swimmers ($P < 0.001$) and the phase differences were much more variable ($P < 0.001$). It is apparent from Figure 2b that there were large differences among participants even among those at the same level of the "learn-to-swim" programme. In general, the knee – ankle phase differences of the swimmers in the learner groups were larger than those of the skilled group ($P < 0.001$) and much more variable ($P < 0.001$) (Figure 2c).

The difference between the velocity index of the skilled and pooled learner groups was great ($P < 0.001$), as was the difference in within-participant variability ($P < 0.05$). All skilled swimmers had an index close to or greater than 1.0, indicating that the phase relationships of the vertical motions of the joints were suited to the task given the rationale in the Introduction (Figure 2d). Conversely, of the learning swimmers, only L1S2 and L3S2 had an index that indicated coordination of vertical motions suited to effective performance of the task. However, in the case of L1S2 in particular, the variability was great.

Biological noise

The differences between the skilled and pooled learner groups in biological noise were large for ankle ($P < 0.001$), knee ($P < 0.05$), and hip ($P < 0.001$). The within-participant variability in biological noise

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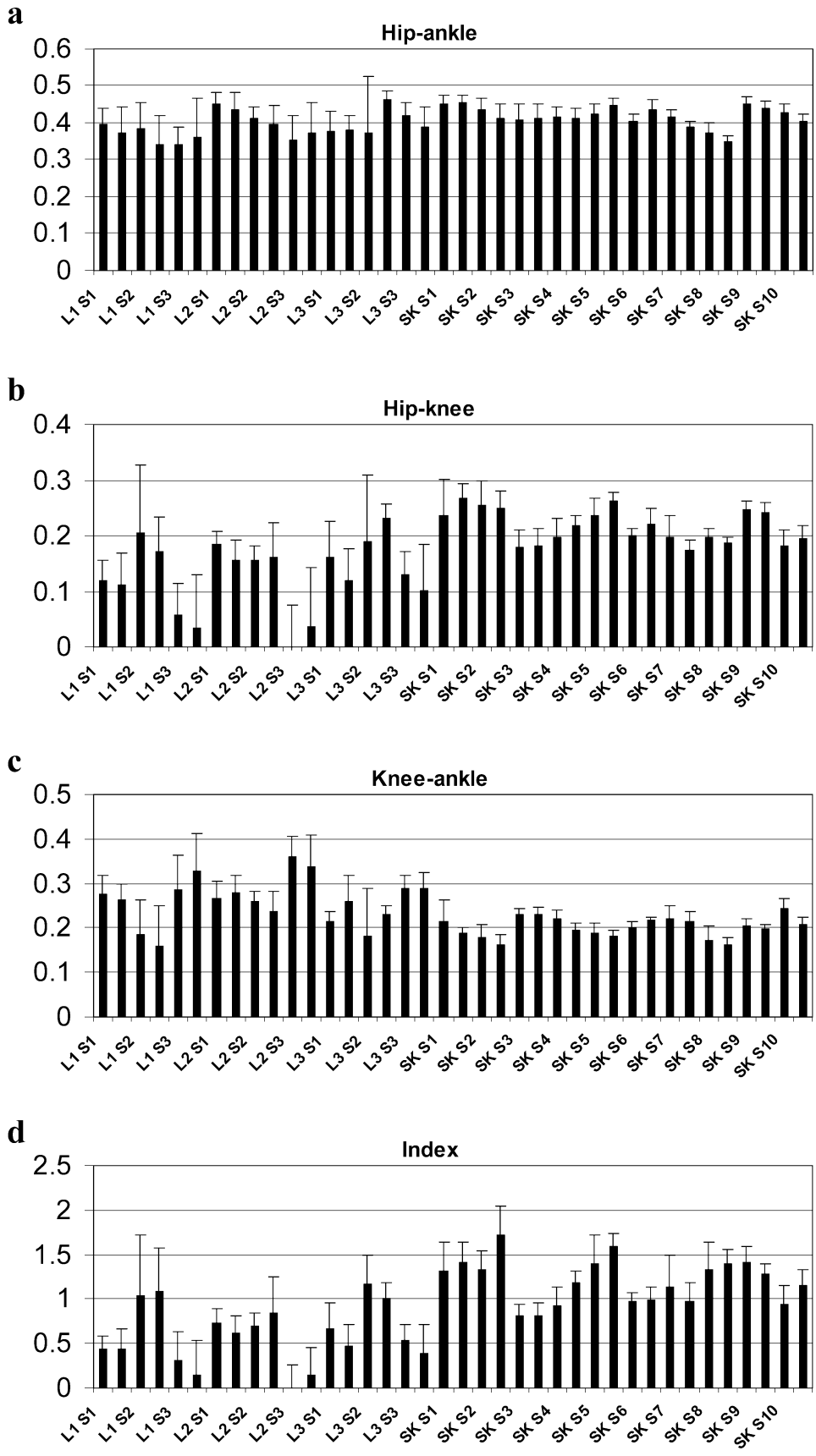

Figure 2. Phase differences (as a proportion of the whole wave cycle) between (a) hip and ankle, (b) hip and knee, and (c) knee and ankle of the fundamental Fourier waveform, and (d) the velocity index (knee – ankle wave velocity/hip – knee wave velocity).

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was significantly greater for the pooled group of learners than for the skilled group for the ankle ($P < 0.01$), knee ($P < 0.01$), and hip ($P < 0.001$). Table V gives the mean biological noise of the combined left and right hip, knee, and ankle joints and the standard deviations for each of the groups. Clearly, Level 1 was much less smooth in the vertical undulations of the hip and knee than the other groups and had irregularities that were assigned to higher frequency waveforms.

The biological noise and its variability for individual participants are shown in Figures 3a, 3b, and 3c for hip, knee, and ankle respectively. Clearly, none of the swimmers in the learning groups had eliminated biological noise from the actions of all the joints as effectively as the skilled swimmers. Furthermore, the level of biological noise across left and right sides and across the three joints was variable within learning groups.

Discussion

With respect to the main purpose of the study – that is, to investigate how the movement patterns of swimmers at different levels of a "learn-to-swim" programme differ from those of skilled performers – several differences are apparent. The pooled learning group and the skilled group were statistically different in the maximum angles of the hip (182.9° and 189.8° respectively) and the ankle (168.1° and 173.3° respectively), and approached significance ($P = 0.077$) in the knee (175.6° and 179.7° respectively). This indicates that the learners were not extending the joints as much as the skilled swimmers. The minimum angles indicated that there was more dorsi flexion in the ankle of the learners (140.0°) than the skilled swimmers (152.9°). Investigation of the means for the three levels of the "learn-to-swim" programme (Table II) reveals that minimum angles in the hip and knee at Levels 1 and 2 were less than those of Level 3 and the skilled swimmers.

Thus, it is apparent that many learners in the early levels of a "learn-to-swim" programme use joint angles that are smaller than those of skilled swimmers. From a hydrodynamic perspective, this disrupts the water flow along the body, thereby increasing resistive drag, as well as reducing the capacity to produce a hydrodynamically efficient "'whip-like" propulsive action. With respect to the hip, this requires a relatively small range of motion with some extension as well as flexion. Similarly, small angles of the knee and ankle and large ranges of motion disrupt the flow in both the hydrodynamic sense and in terms of the flow of motion in a "whip-like" manner.

The results are interesting in the light of the idea that development of kicking patterns will be influenced by existing movement patterns (Thelen, 1995). For example, the angles of the ankle joint used in the early levels of the "learn-to-swim" programme may be influenced by the activities with which they are most familiar, in particular walking. This may contribute to the tendency to have a considerable range of ankle motion but with too much dorsi flexion and inadequate plantar flexion for the task of flutter kicking.

Implications for teaching flutter kicking emerge from the findings. Teachers can encourage beginners to constrain their hip flexion and to release any constraint that limits their hip extension. A common cause of limited hip extension is excessive knee flexion. This may be related to a beginner's perception of the task. A beginner may perceive the task as one of "kicking from the knee". When the knee is flexed to accommodate this perception, the foot comes high out of the water unless the hip is flexed. To keep the leg below the surface of the water, the swimmer constrains hip extension. Therefore, releasing the constraint that limits hip extension may require that the learner's perception of the task is changed via instruction. Similarly, to "override" the ranges of motion habitually used in other activities, the teacher can simply encourage the learner to "point the toes".

Differences in coordination between the pooled learner group and the skilled group were manifest in significant differences in the timing of hip, knee, and ankle minima and maxima with the only exception being the time of attainment of maximum ankle angle (Table III). Coordination differences were also evident in the Fourier phase differences for the hip to knee and knee to ankle (Figure 2). As a consequence, the wave velocity from hip to knee was significantly larger in the learner group than the skilled group (Table IV) and the velocity index was smaller (Figure 2d).

The results indicated a hydrodynamically efficient "whip-like" coordination among the skilled swimmers. While the phase difference of vertical motion between ankle and knee was typically appropriate for performance of the task among learners, the hip-to-knee phase difference was too small for most of the learners. This meant that the hip and knee undulations were almost simultaneous and incompatible with the goal of establishing a body wave that

Table V. Mean biological noise (%) hip, knee, and ankle joints with standard deviations (in parenthesis).

	L1		L2		L3		Skilled	
Hip	19.6	(18.3)	7.0	(6.7)	5.9	(5.0)	2.3	(1.8)
Knee	14.2	(16.8)	3.5	(3.7)	4.2	(5.8)	2.8	(1.3)
Ankle	1.9	(1.8)	1.4	(1.4)	1.9	(2.4)	0.6	(0.3)

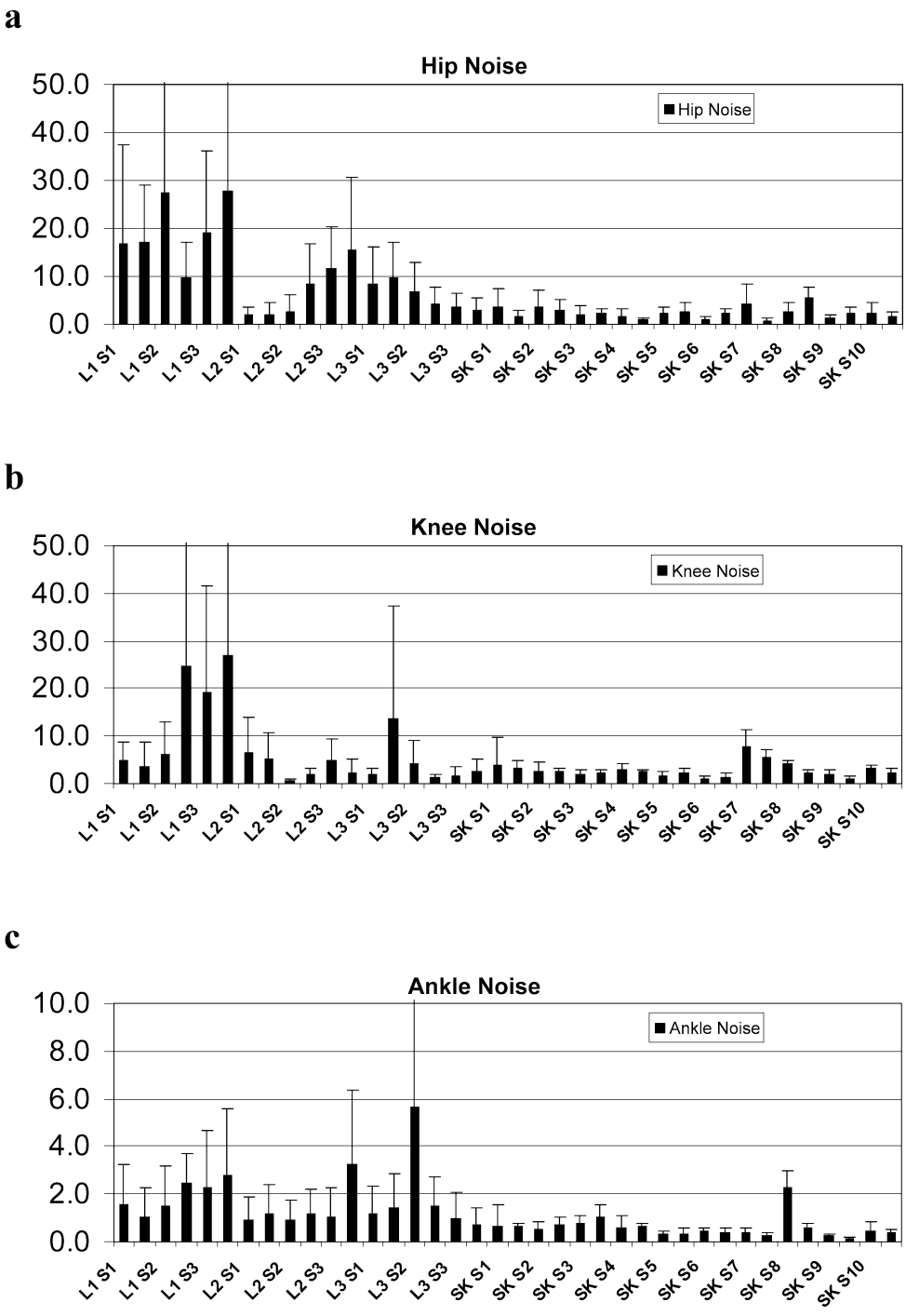

Figure 3. Biological noise (%) for (a) hip, (b) knee, and (c) ankle.

does not decrease, and preferably increases in velocity as it moves caudally. Therefore, teachers have a role to play in encouraging the development of a sequential coordination of joint undulations to promote a body wave.

The data reported here provide a good example of the commonly observed phenomenon of convergence. The data show clearly that learning swimmers can differ greatly from each other in many variables in the flutter kick, including joint angles, ranges of motion, and timing. For example, while it is common for learning swimmers to have simultaneous rather than sequential hip and knee actions, a swimmer at the first level of the "learn-to-swim" programme (L1S2) had an appropriate phase difference in hip and knee actions and this was reflected in an index that was similar to many of the skilled swimmers, although considerably more variable.

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Conversely, a swimmer in the second level of the "learn-to-swim" programme (L2S3) still had inappropriate simultaneous timing. Thus, while there are some common differences from skilled swimmers, teachers should be aware that individuals vary greatly in their characteristics even at similar levels of a "learn-to-swim" programme.

Also, teachers can expect inconsistency in the early levels of a "learn-to-swim" programme. In fact, in terms of dynamical systems theory (Kelso *et al.*, 1981; Kugler *et al.*, 1980), variability may be viewed as desirable and functional as it ensures that the learner's system has adequate "experience" to develop "attractor states" that are appropriate to meet the goal of efficient performance (Davids, Glazier, Araujo, & Bartlett, 2003). From this perspective, it may not be prudent to be excessively conscientious in guiding learners towards the skilled model before they have had a chance to explore fully the movement possibilities.

On the other hand, teachers must ensure that the development of inappropriate movement patterns is avoided. This is where a knowledge of which kinematic features are characteristic of skilled performance can be very useful. In this study, it was shown that in addition to large within-participant variability, learners displayed great between-participant variability and also exhibited some differences from skilled performance that were common. In cases where individual differences are clearly outside the range of movement patterns of skilled performers, learners can be guided towards the appropriate movement patterns. This may involve "shaping" the movement by introducing constraints or by removing existing unhelpful constraints. The challenge for the instructor is to find the right balance between facilitating exploration of movement possibilities and guiding the learner towards movement patterns that are known to be effective, or away from movement patterns that are known to be ineffective.

Among the learning groups, only L1S2 and L3S2 showed appropriate coordination, as indicated by the index scores (Figure 2d). However, the high variability of these participants indicated that the coordination was far from established. This supplements other empirical evidence (e.g. Swinnen, Walter, & Shapiro, 1988) indicating that, even if an appropriate coordination is discovered early in the learning of a task, considerable practice is required to stabilize it. While the coordination of L3S2 was less variable than that of L1S2, indicating the desired movement pattern had been established more strongly, it was more variable than that of most of the skilled swimmers. In particular, the hip – knee coordination (Figure 2b) was variable.

The fact that participants in the third level of a "learn-to-swim" programme (L3S1 and L3S2) were still struggling to move away from the simultaneous motion of hip and knee towards the more appropriate sequential phase relationship is interesting in the light of the development of kicking actions whereby strong movement patterns, or in dynamical systems terms "stable attractors", for kicking are established at an early age (Thelen, 1995). Thelen reported that the kicking movement patterns of infants were very similar to their stepping movement pattern. The flutter kicking task was clearly very different from previously established and stable kicking patterns. Thus, a pattern appropriate for propulsion in swimming had not been established consistently among learners in the three levels of the "learn-to-swim" programme. From an instructional perspective, teachers need to be aware that progression to a level of a "learn-to-swim" programme in which swimmers would be regarded as competent does not necessarily imply that relevant sub-skills are being performed with appropriate coordination. In this case, two of the three swimmers at the third level of the "learn-to-swim" programme had not developed the desirable sequencing of hip and knee undulation. While this process may be largely self-pacing and related to the amount of practice or to developmental aspects, it is possible that the process can be accelerated by influencing the learner's perception of the task. If the task is perceived as one in which the motion should "flow" from the hip to the knee to the ankle then the development of effective coordination may be more rapid than if the learner has the perception that the task simply involves moving the foot up and down. Thus, a teacher might improve instructional effectiveness by encouraging the development of appropriate mental images using analogies such as "cracking a whip".

The kicks of the skilled group had significantly less noise than those of the pooled learning groups for the hip, knee, and ankle vertical undulations. This result is not surprising given that a single sinusoidal vertical undulation of joints that progresses caudally is associated with efficient swimming (Videler, 1981).

This study has yielded interesting data relating to contemporary motor control issues. Skilled swimmers were distinguished from the learning swimmers by three main indicators: (1) they sequenced hip – knee and knee – ankle vertical motions to create a body wave whereby the index of knee – ankle to hip – knee velocity was close to or greater than 1.0; (2) they showed low variability in all the kinematic measures across the hip, knee, and ankle joints; and (3) their motions were smooth as indicated by small percentages of biological noise. While some of the learning swimmers had attained similar values to skilled

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swimmers in some of the variables, none had achieved comparable values across all three indicators across all three joints. This suggested that all of the learning swimmers were still "exploring" the movement possibilities, or in dynamical systems terms exploring the "state space" to develop appropriate movement patterns. The three indicators were independent of each other, and the data indicated that individuals were "fine-tuning" each independently. For example, L3S2 had an appropriate index, but with more variability than skilled swimmers, a large amount of biological noise in the left ankle, and much less biological noise in the right ankle and in both sides of the knee and hip. In contrast, L3S3, despite being at the same level of a "learn-to-swim" programme, had small variability and small biological noise levels, but had not developed the appropriate sequencing.

Further studies using longitudinal designs would provide a more complete picture of how different swimmers progress and "fine-tune" each of the task-related indices characteristic of skilled swimmers. It may be, for example, that L3S3 developed a stable "attractor state" that allowed reduction in variability and elimination of noise. Further improvement may involve destabilizing that coordination to search for a better coordination – that is, one closer to the desired body wave velocity index. The data are congruent with the view that learning involves a series of states of stability, instability, and phase shifts (Thelen, 1995).

The common finding that learners converge towards characteristics common to skilled performers was evident in this study with respect to task-relevant variables such as joint angles, ranges of motion, and timing of actions. However, one aspect of interest in this particular study is that skilled performance of the flutter kick requires development of a simple movement pattern – that is, a simple sinusoidal vertical undulation of each joint. Higher frequency components were undesirable for task achievement and had been eliminated by skilled swimmers. This is not to say that the coordination of the joint rotations to achieve that is simple. However, the sought movement topology is simple. This is in contrast to studies of other tasks in which Fourier analysis was applied to investigate convergence of movement patterns towards a topology suited to task achievement. Marteniuk and Romanow (1983) found that participants operating a manipulandum incorporated higher frequency harmonics when attempting to reproduce a complex displacement-time waveform. Sanders and Wilson (1992) found that participants jumping from a sprung surface progressively adjusted the pattern of acceleration relative to the sprung surface by increasing the amplitude of the higher frequency odd harmonics.

The joint angular data provided no evidence that degrees of freedom were "frozen" in the early levels of a "learn-to-swim" programme. Rather, it appeared that all learners at all stages were exploring the movement possibilities. This was indicated by large variability and ranges of joint motion. In fact, many of the learners had ranges of joint motion greater than those of the skilled swimmers. Thus, the findings for the flutter kick were somewhat similar in this regard to the findings of Ko *et al.* (2003) for the dynamic balance task and add further support to the statement by Newell and Vaillancourt (2001) that the sequence of freezing and freeing degrees of freedom is task-dependent.

Furthermore, similar to the findings of Ko *et al.* (2003), there was evidence of coupling between joints. In particular, there was a predominance of positive correlations among the skilled swimmers and many of those correlations were strong for knee – hip joint motion. Although not statistically conclusive the results indicated that there was "something happening" that justifies further investigation to determine how the complexity of rhythmical multi-joint tasks is reduced with increasing skill.

Conclusion

Kicking actions of skilled swimmers are characterized by a task-related sequencing of hip, knee, and ankle vertical undulations that differs from that of less skilled swimmers. Swimmers in the early levels of a "learn-to-swim" programme have a more simultaneous coordination of hip and knee actions than skilled swimmers. Skilled swimmers also have less variability and less noise than swimmers at early levels of a "learn-to-swim" programme. Common differences in kinematics between beginning swimmers and skilled swimmers in prone flutter kick include excessive hip flexion and inadequate hip extension, excessive knee flexion, and inadequate plantar flexion of the ankle. The results imply that teachers might improve teaching effectiveness by guiding swimmers towards appropriate ranges of motion and indicating to the learner the boundaries in the ranges of motion that should be used. Teachers might also assist learning by emphasizing the need to initiate the kick from the hip so that the motion "flows" to the feet rather than simply moving the feet up and down. The data suggest that in learning the flutter kick swimmers converged towards appropriate ranges of motion and coordination of joint actions to yield simple sinusoidal movement topologies related to task achievement. In doing so, biological noise was reduced. There was some evidence that skilled swimmers "froze" degrees of freedom by coupling the range of joint motion between joints.

Acknowledgement

This study was conducted as part of the Swimming Nature Limited Research and Development Programme. Swimming Nature Limited provided funding to support a student, Kelly Myers, to assist with data collection and processing. Thanks are expressed to the CEO of Swimming Nature Limited, Eduardo Ferre, for recognizing the need for research to advance methods of teaching swimming.

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